

ReGraNet: A Retinal Graph Reasoning Network for Diabetic Retinopathy Grading

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Abstract—Diabetic Retinopathy (DR) remains one of the leading causes of preventable blindness worldwide, making reliable and early-stage screening essential for timely clinical intervention. While convolutional neural networks (CNNs) have demonstrated strong performance in retinal image analysis, they primarily emphasize local texture patterns and often underutilize anatomical vessel structure and long-range contextual dependencies. To address these limitations, this paper proposes ReGraNet, a hybrid deep learning framework that jointly integrates CNN, Graph Neural Network (GNN), and Transformer branches for comprehensive DR grading. The CNN branch captures lesion-aware multi-scale visual representations, the GNN branch models retinal vascular connectivity through graph-based reasoning, and the Transformer branch learns global relationships via patch-wise self-attention. Furthermore, a bidirectional cross-attention bridge is introduced to facilitate mutual interaction between structural graph embeddings and contextual image tokens. The fused multimodal representation is processed by a multilayer perceptron for five-class DR severity prediction. Experimental results on benchmark retinal datasets demonstrate that the proposed framework achieves strong classification performance, robust generalization, and improved interpretability. In addition, Grad-CAM and attention visualizations reveal clinically meaningful decision regions, highlighting the potential of multimodal retinal reasoning for reliable and explainable automated screening systems.

I. INTRODUCTION

Diabetic Retinopathy (DR) is a progressive microvascular complication of diabetes mellitus and remains one of the primary causes of preventable blindness worldwide [1], [2]. As the global prevalence of diabetes continues to increase, the burden of DR is expected to rise significantly, creating substantial clinical and socioeconomic challenges [3]. Since vision loss caused by advanced retinal damage is often irreversible, early diagnosis and timely treatment are critical for preserving visual function and improving patient outcomes [4].

In clinical practice, DR screening is commonly performed using color fundus photography, where ophthalmologists assess retinal images for pathological signs such as microaneurysms, hemorrhages, hard exudates, cotton wool spots, venous abnormalities, and neovascularization [5], [6]. However, large-scale manual screening is labor-intensive, time-consuming, and dependent on the availability of experienced specialists [7]. These limitations are particularly severe in low-

resource and rural settings, where access to ophthalmic expertise is often limited [8]. Consequently, automated computer-aided diagnosis systems have emerged as an important research direction for scalable and cost-effective retinal screening [9].

Recent advances in deep learning have significantly improved automated retinal image analysis [10]. In particular, Convolutional Neural Networks (CNNs) have achieved strong performance in DR classification by learning hierarchical spatial representations directly from fundus images [11], [12]. CNN-based methods are highly effective in detecting local lesion patterns, texture irregularities, and fine-grained visual abnormalities [13]. Nevertheless, conventional CNNs operate on regular grid structures and may not explicitly capture the topological organization of retinal vasculature, including vessel branching patterns, junction connectivity, and structural abnormalities that are clinically relevant indicators of disease progression [14].

More recently, Transformer-based architectures have demonstrated remarkable capability in modeling long-range dependencies through self-attention mechanisms [15], [16]. By processing images as sequences of patches, Vision Transformers can capture global contextual relationships between distant retinal regions [17]. Despite this advantage, Transformer models do not inherently encode graph-structured anatomical priors such as vascular networks [18]. In contrast, Graph Neural Networks (GNNs) are well suited for learning from non-Euclidean relational data and can effectively model retinal vessel topology when vessels are represented as graphs [19], [20]. However, graph-only approaches may lack rich appearance information available in pixel space [21].

These observations suggest that local lesion characteristics, global contextual interactions, and vascular structural priors provide complementary information for DR grading [13], [22]. Motivated by this insight, we propose **ReGraNet** (Retinal Graph Reasoning Network), a unified hybrid framework that jointly leverages CNNs, GNNs, and Transformers for robust diabetic retinopathy classification. Specifically, the CNN branch extracts lesion-aware multi-scale visual features, the GNN branch reasons over vessel junction connectivity using graph attention learning [23], and the Transformer branch models global retinal context through patch-wise self-attention [16]. To further strengthen feature interaction, a

bidirectional cross-attention bridge is introduced between the graph and Transformer branches, enabling mutual refinement of structural and contextual representations [24]. The resulting multimodal embeddings are fused through a lightweight classification head for five-class DR severity prediction.

The main contributions of this work are summarized as follows:

- We propose a novel hybrid architecture that integrates CNN, GNN, and Transformer branches for comprehensive diabetic retinopathy grading.
- We construct retinal vessel graphs from fundus images to explicitly encode vascular topology, branching patterns, and anatomical connectivity [14], [20].
- We introduce a bidirectional cross-attention mechanism to enhance interaction between graph-based structural features and global contextual representations [24].
- We incorporate explainability analysis using Grad-CAM [25] and attention visualization to improve clinical interpretability and trustworthiness.

II. RELATED WORK

Automated diabetic retinopathy grading has been extensively studied using computer vision and deep learning techniques. This section reviews the most relevant prior work organized into five categories: traditional image processing methods, CNN-based approaches, Transformer-based methods, Graph Neural Network approaches, and multimodal fusion strategies.

A. Traditional and Handcrafted Methods

Early approaches to automated DR detection primarily relied on handcrafted image processing pipelines involving lesion segmentation, vessel extraction, texture descriptors, and manually designed classifiers [36], [37]. Techniques such as matched filtering, morphological operations, and Gaussian mixture models were used to extract retinal features including blood vessels, optic disc boundaries, and lesion regions [38]. Feature engineering methods combining local binary patterns (LBP), histogram of oriented gradients (HOG), and support vector machines (SVMs) were also widely applied for DR classification [39], [40]. Although these methods provided clinically interpretable cues and required relatively modest computational resources, their performance was often constrained by limited generalization across imaging devices, illumination variations, and dataset heterogeneity [41]. The dependency on domain expertise for feature design further restricted their scalability in large-scale screening programs [7].

B. CNN-Based Approaches

The emergence of deep convolutional networks significantly advanced retinal disease analysis by enabling end-to-end feature learning directly from fundus images [10]. Architectures such as ResNet [42], DenseNet [43], InceptionNet [44], and EfficientNet [45] have demonstrated strong performance in detecting diabetic retinopathy severity by learning hierarchical

spatial representations. Gulshan et al. [9] demonstrated that a deep CNN trained on a large clinical dataset could achieve diagnostic performance comparable to ophthalmologists in detecting referable DR. Subsequent work by Gargeya and Leng [12] further showed that CNN-based models could generalize across diverse clinical datasets without hand-engineered features. CNN-based models are particularly effective in identifying local pathological patterns such as microaneurysms, hemorrhages, and exudates [11], [46]. Zoom-in networks and multi-scale lesion detection frameworks were proposed to capture fine-grained retinal abnormalities at varying resolutions [47]. Attention-guided CNNs have also been explored to weight discriminative retinal regions during feature extraction [48]. However, because convolution operates over local receptive fields, these approaches may underutilize long-range dependencies and global structural context that are relevant to DR staging [13].

C. Transformer-Based Approaches

Transformer-based models have recently gained significant attention in medical image analysis due to their self-attention mechanism, which can model interactions between distant image regions [15]. Dosovitskiy et al. [16] introduced the Vision Transformer (ViT), which divides images into fixed-size patches and processes them as token sequences, enabling global receptive fields from the first layer. Hybrid CNN-Transformer architectures such as TransUNet [18] and Swin Transformer [49] have shown competitive results for retinal image segmentation and classification by combining local convolutional features with global attention. Sun et al. [50] proposed a lesion-aware Transformer for DR grading that incorporates lesion location priors into the attention computation, improving sensitivity to small pathological regions. Despite these strengths, Transformer models typically require large-scale training data and exhibit high computational complexity during training [16]. Crucially, they do not explicitly encode anatomical graph priors such as vessel connectivity or branching topology, which are established clinical markers of DR progression [18].

D. Graph Neural Network-Based Approaches

Graph Neural Networks (GNNs) provide a powerful alternative for learning from relational and non-Euclidean data structures [19], [23]. In retinal analysis, blood vessel networks can naturally be represented as graphs where bifurcation points and junctions serve as nodes, while vessel segments define graph edges [14], [20]. By propagating information through connected nodes, GNNs can capture anatomical topology, structural irregularities, and connectivity changes associated with disease progression [21]. Meng et al. [51] proposed a graph convolutional approach for optic disc and cup segmentation by modeling spatial relationships between retinal landmarks. Graph attention networks (GATs) [23] have been applied to model vessel junction features with adaptive neighbor weighting, improving robustness to imaging noise and topological variation. Chen et al. [52] demonstrated that

incorporating vascular graph structure as a prior significantly improves DR grading compared to pixel-only features. Geometric deep learning frameworks have also been used to analyze retinal vascular morphology and detect early biomarkers of microvascular disease [53]. Nevertheless, graph-based methods alone may miss discriminative pixel-level appearance features that remain crucial for lesion recognition, motivating hybrid architectures [21].

E. Multimodal and Hybrid Fusion Approaches

Several recent studies have explored multimodal fusion strategies combining convolutional and attention-based learning to improve DR grading performance [13], [22]. Zhou et al. [13] proposed a collaborative learning framework combining segmentation and classification branches to jointly optimize lesion detection and severity grading. Sun et al. [50] integrated lesion detection priors with global image context through a dual-branch architecture. He et al. [54] proposed a category attention block combined with multi-scale feature aggregation to capture inter-class relationships for fundus disease classification. Lu et al. [24] introduced bidirectional cross-attention between modalities in a visual-linguistic setting, motivating its adaptation for cross-modal medical image fusion. While such methods improve representation diversity, most existing frameworks still focus on image-space features and rarely incorporate explicit vascular graph reasoning in a unified architecture [22]. Furthermore, limited work has investigated structured interaction between graph embeddings and global contextual tokens through dedicated cross-modal attention mechanisms, leaving an important gap in current literature.

III. PROPOSED METHODOLOGY

This section presents the proposed **ReGraNet** framework for automated five-class diabetic retinopathy grading. The central motivation of the framework is that retinal disease severity is characterized not only by local lesion appearance, but also by vascular structural changes and global contextual relationships across the fundus image [4], [9]. To exploit these complementary cues, ReGraNet jointly combines CNN, GNN, and Transformer branches within a unified multimodal architecture [13], [22]. The overall pipeline consists of image preprocessing, retinal vessel graph construction, branch-wise feature learning, cross-modal interaction, multimodal fusion, and final classification.

A. Image Preprocessing

Fundus images collected from different imaging devices often exhibit significant variations in illumination, contrast, resolution, and background artifacts, which can adversely affect downstream feature learning if left unaddressed [7], [38]. To reduce such variability and ensure consistent input representation, a standardized preprocessing pipeline is applied to all images prior to model training and inference.

Each image is first resized to 512×512 pixels to obtain a consistent input resolution suitable for multi-scale feature

extraction [45]. Contrast Limited Adaptive Histogram Equalization (CLAHE) is then applied to enhance local contrast and improve visibility of subtle lesions and thin vessel structures that may otherwise be obscured in regions of uneven illumination [55]. CLAHE operates on localized image tiles rather than the global histogram, preventing over-amplification of noise in regions with low signal variation, which is particularly beneficial for detecting microaneurysms and small hemorrhages in dark retinal areas [56].

To further standardize input across heterogeneous imaging sources, per-channel mean subtraction and standard deviation normalization are applied using statistics computed from the training set [42]. This normalization improves gradient flow during training and accelerates convergence, especially for deep backbone networks pretrained on natural image datasets.

During training, data augmentation strategies are employed to improve generalization and reduce overfitting [57]. Specifically, random horizontal and vertical flipping, rotation within ± 30 degrees, random brightness adjustment, contrast perturbation, and color jitter are applied stochastically. These transformations expose the model to diverse imaging variations and simulate differences in camera orientation, patient positioning, and acquisition settings encountered across clinical sites [58]. Augmentation strategies are applied on-the-fly during training to maximize effective data diversity without requiring pre-generation of augmented samples.

B. Retinal Vessel Graph Construction

Retinal vasculature contains clinically meaningful structural information, including tortuosity, branching density, vessel continuity, and abnormal neovascular patterns, all of which are important biomarkers of diabetic retinopathy severity [4], [14]. To explicitly encode this anatomical structure as a learnable representation, a retinal vessel graph is constructed from the preprocessed fundus image prior to GNN processing.

The vessel graph construction pipeline proceeds through three main stages: vessel segmentation, skeleton extraction, and graph encoding. First, a vessel mask is extracted from the preprocessed fundus image using a combination of Frangi vesselness filtering [59] and morphological operations including erosion and dilation. Frangi filtering enhances tubular structures by analyzing the eigenvalues of the Hessian matrix at multiple scales, making it well suited for detecting vessels of varying widths across the retinal image. The resulting binary vessel map delineates the full vascular network present in the fundus.

Following vessel segmentation, morphological skeletonization is applied to reduce the vessel mask to a one-pixel-wide medial axis representation [60]. This thinning operation preserves the topological connectivity of the vessel network while removing redundant width information. Junction points and branching intersections are then identified from the skeleton by analyzing local pixel neighborhoods: a skeleton pixel is classified as a junction if it has three or more connected neighbors. These detected junctions and anatomically salient endpoints are treated as graph nodes.



Fig. 1. Preprocessing pipeline for retinal fundus analysis: (a) original fundus image, (b) CLAHE-enhanced image improving local contrast and highlighting fine vascular structures, and (c) corresponding binary vessel segmentation mask delineating the retinal vasculature for downstream feature extraction and analysis.

Let the set of detected nodes be denoted as $V = \{v_1, v_2, \dots, v_n\}$, where each node v_i is represented by its normalized spatial coordinates (x_i, y_i) within the image grid. Additional node features such as local vessel orientation, branch degree, and neighborhood vessel density statistics derived from the vessel map are also incorporated to enrich node representations [20]. A k -nearest neighbor (k-NN) strategy is used to connect each node to its k spatially closest neighbors, thereby constructing graph edges that approximate vessel connectivity:

$$E = \{(v_i, v_j) \mid v_j \in \mathcal{N}_k(v_i)\},$$

where $\mathcal{N}_k(v_i)$ denotes the set of k nearest neighbors of node v_i . The graph is formally defined as $G = (V, E)$. The resulting graph provides a compact non-Euclidean representation of retinal topology for subsequent graph reasoning [19], [53].

This graph construction approach offers several advantages over pixel-level vessel representations. It reduces the spatial complexity of vascular data to a compact relational structure, making downstream graph learning computationally efficient. Moreover, by encoding vessel junction connectivity explicitly, the graph captures network-level properties such as loop formation, branching asymmetry, and capillary dropout that are difficult to extract from unstructured image feature maps [52].

C. CNN Branch: Local Lesion Representation

The CNN branch is responsible for extracting fine-grained pathological patterns that are strongly associated with diabetic retinopathy progression [9], [10]. Typical retinal lesions such as microaneurysms, dot-blot hemorrhages, hard exudates, cotton wool spots, and texture distortions are often localized and vary significantly in size, contrast, and shape [5]. Therefore, a convolutional backbone is employed to learn hierarchical spatial features directly from pixel space, leveraging the inductive bias of local connectivity and weight sharing that makes CNNs well suited to structured visual data [42].

In this work, EfficientNet-B4 [45] is selected as the backbone network due to its compound scaling strategy, which jointly balances network depth, width, and input resolution for an improved accuracy-efficiency trade-off. Compared to manually designed architectures, EfficientNet achieves competitive performance with fewer parameters, making it suitable for medical imaging tasks where training data may be limited [45]. The backbone is initialized with ImageNet pretrained weights, and fine-tuned on the retinal training set to leverage low-level visual features learned from large-scale natural images [61]. The early convolutional layers capture low-level cues such as edges, color transitions, vessel boundaries, and illumination changes, while deeper layers learn higher-level semantic lesion representations that are more specific to retinal pathology [62].

To preserve discriminative information across multiple spatial scales, intermediate feature maps from different backbone stages are aggregated using a feature pyramid style fusion mechanism [63]. This is particularly important in retinal analysis because very small lesions (e.g., microaneurysms of a few pixels in diameter) and larger abnormalities (e.g., hemorrhagic regions spanning significant retinal area) coexist within the same image [47]. Feature Pyramid Network-style fusion upsamples deeper semantic features and combines them with shallower spatially detailed features through lateral connections and element-wise addition, enabling the model to remain sensitive to both subtle and coarse disease indicators [63].

Formally, let $\mathbf{X} \in \mathbb{R}^{H \times W \times 3}$ denote the input image. The CNN encoder transforms the image into a hierarchy of feature maps:

$$\{\mathbf{F}_1, \mathbf{F}_2, \dots, \mathbf{F}_L\} = \Phi_{cnn}(\mathbf{X}),$$

where Φ_{cnn} denotes the EfficientNet backbone and L is the number of selected stages. These features are fused and globally pooled to produce a compact lesion-aware representation:

$$\mathbf{f}_{cnn} = \text{GAP}(\text{Fuse}(\mathbf{F}_1, \mathbf{F}_2, \dots, \mathbf{F}_L)),$$

where GAP denotes global average pooling [64]. The resulting embedding $\mathbf{f}_{cnn} \in \mathbb{R}^{d_{cnn}}$ provides robust local visual evidence for downstream classification.

In addition to improved representation quality, the CNN branch also contributes inductive bias for spatial locality, making it particularly effective in scenarios with limited training data where purely attention-based models may overfit [16]. The combination of pretrained initialization, multi-scale aggregation, and global pooling makes this branch a strong foundation for local lesion characterization within the multi-modal ReGraNet framework.

D. GNN Branch: Structural Topology Reasoning

Retinal vasculature undergoes significant structural changes during the progression of diabetic retinopathy [4], [20]. Clinical manifestations such as vessel narrowing, abnormal branching, tortuosity, capillary dropout, neovascularization, and irregular connectivity patterns often provide valuable diagnostic evidence beyond pixel-level lesion appearance [5], [14]. Since such vascular relationships are inherently non-Euclidean, representing the vessel network as a graph offers a natural and effective learning paradigm [19], [53].

In the proposed framework, the extracted vessel skeleton is converted into a graph $G = (V, E)$, where each node $v_i \in V$ corresponds to a vessel junction, bifurcation point, or salient landmark, and each edge $(v_i, v_j) \in E$ represents spatial connectivity between neighboring vessel points [20]. Node features may include normalized coordinates, local vessel orientation, branch degree, and neighborhood statistics derived from the vessel map [52].

Unlike conventional CNNs that operate on regular image grids, Graph Neural Networks perform message passing directly over graph connectivity, allowing information to propagate through anatomically meaningful vessel pathways [19], [23]. This enables the model to capture structural dependencies that are difficult to encode in Euclidean feature spaces and allows the network to reason about vascular topology at multiple levels of abstraction [53].

To learn topology-aware representations, a multi-layer Graph Attention Network (GAT) [23] is employed. For each node i , attention coefficients are computed over its neighboring nodes to determine their relative importance:

$$e_{ij} = \text{LeakyReLU}(\mathbf{a}^T [\mathbf{W}\mathbf{h}_i \parallel \mathbf{W}\mathbf{h}_j]),$$

where $\mathbf{h}_i$ and $\mathbf{h}_j$ are node features, $\mathbf{W}$ is a learnable projection matrix, $\mathbf{a}$ is the attention vector, and $\parallel$ denotes concatenation [23]. The coefficients are normalized using softmax:

$$\alpha_{ij} = \frac{\exp(e_{ij})}{\sum_{k \in \mathcal{N}(i)} \exp(e_{ik})}.$$

The updated node representation is then obtained as:

$$\mathbf{h}'_i = \sigma \left(\sum_{j \in \mathcal{N}(i)} \alpha_{ij} \mathbf{W}\mathbf{h}_j \right),$$

where $\sigma(\cdot)$ denotes a nonlinear activation function [23]. Through stacked graph attention layers, each node progressively aggregates information from larger vessel neighborhoods, enabling the network to model both local branching behavior and higher-order vascular organization [51].

To enhance representation diversity, multi-head graph attention is employed, where several independent attention heads learn complementary structural patterns [23]. The outputs of multiple heads are concatenated or averaged depending on the layer position:

$$\mathbf{h}'_i = \parallel_{m=1}^M \sigma \left(\sum_{j \in \mathcal{N}(i)} \alpha_{ij}^{(m)} \mathbf{W}^{(m)} \mathbf{h}_j \right),$$

where M denotes the number of attention heads and $\alpha_{ij}^{(m)}$ are the normalized attention coefficients for head m [23]. This is particularly beneficial in retinal graphs because different heads may focus on distinct cues such as dense branching regions, elongated vessel chains, or abnormal peripheral connectivity [52].

After message passing, node-level features are aggregated through global pooling to obtain a fixed-length graph descriptor:

$$\mathbf{f}_{gnn} = \text{Pool}(\{\mathbf{h}'_1, \mathbf{h}'_2, \dots, \mathbf{h}'_n\}),$$

where Pool may denote mean pooling, max pooling, or attention-based graph readout [65]. The resulting embedding $\mathbf{f}_{gnn} \in \mathbb{R}^{d_{gnn}}$ summarizes the global vascular topology of the retina.

The GNN branch contributes three key advantages to the overall framework. First, it explicitly encodes anatomical structure rather than relying solely on appearance cues [53]. Second, it is robust to local intensity variations because graph connectivity is derived from vessel geometry [20]. Third, it complements the CNN and Transformer branches by providing disease-relevant structural priors that improve grading performance and interpretability [51], [52].

E. Transformer Branch: Global Context Modeling

While CNNs are highly effective for local feature extraction, retinal disease grading also depends on global contextual relationships distributed across distant regions of the fundus image [15], [16]. For example, the joint presence of lesions in multiple quadrants, large-scale vascular imbalance, and widespread pathological spread may be more informative than isolated local findings [50]. To capture such long-range dependencies, a Transformer branch is incorporated into the proposed framework.

The preprocessed image is first partitioned into a sequence of non-overlapping patches of size $P \times P$, following the Vision Transformer formulation [16]. Each patch is flattened and projected into a latent embedding space through a learnable linear layer. If the total number of patches is $N = HW/P^2$, the resulting token sequence is:

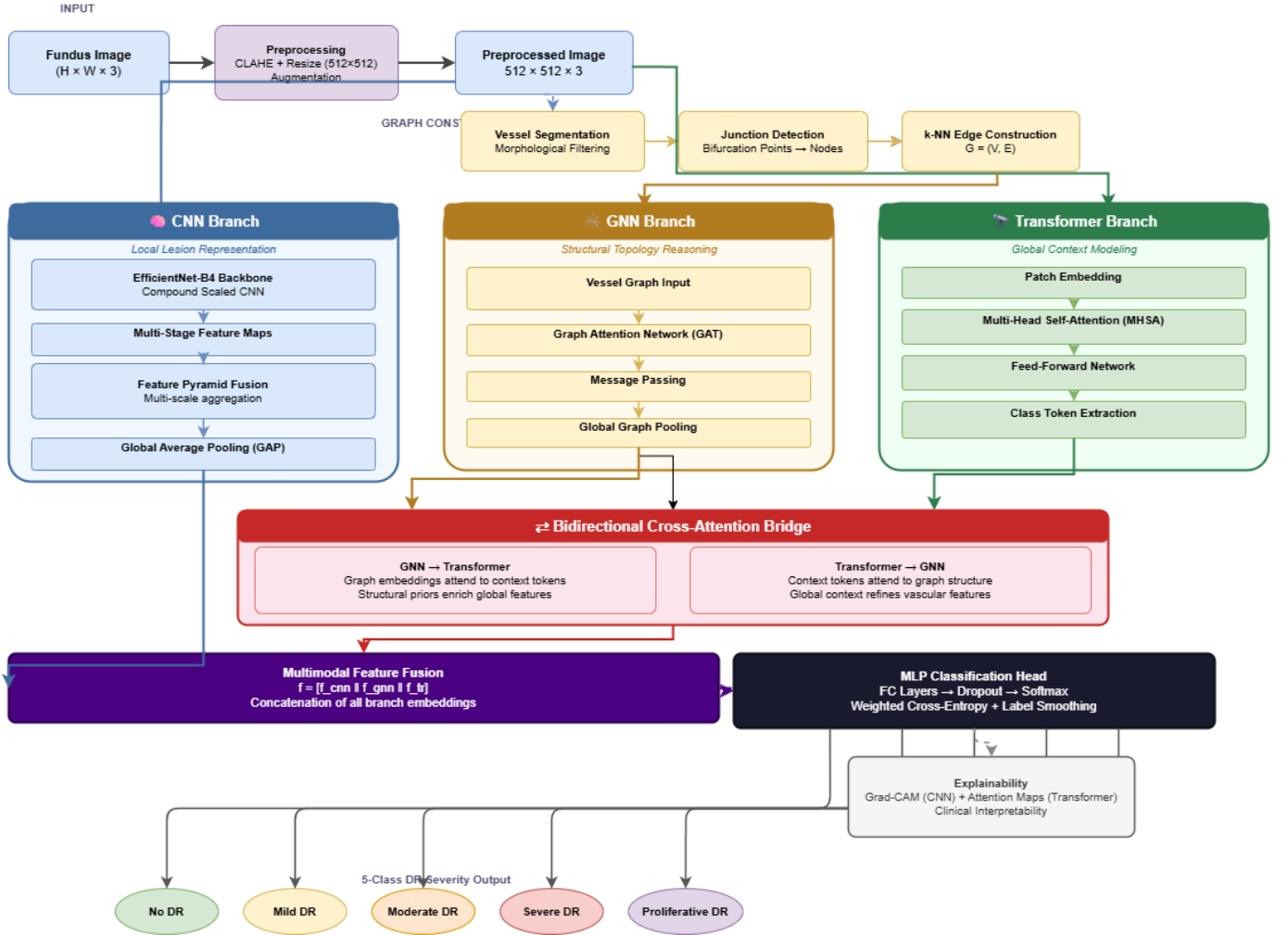


Fig. 2. Overall architecture of the proposed ReGraNet framework. The pipeline processes retinal fundus images through preprocessing and vessel graph construction, followed by parallel CNN, GNN, and Transformer branches for local feature extraction, structural topology reasoning, and global context modeling, respectively. A bidirectional cross-attention mechanism enables inter-branch feature interaction, and the fused representations are passed to an MLP classifier for five-class diabetic retinopathy grading.

$$\mathbf{Z}_0 = [\mathbf{x}_{cls}; \mathbf{x}_p^1; \mathbf{x}_p^2; \dots; \mathbf{x}_p^N] + \mathbf{E}_{pos},$$

where $\mathbf{x}_{cls}$ is the learnable class token prepended to aggregate global image information [16], $\mathbf{x}_p^i$ denotes the embedding of the i -th image patch, and $\mathbf{E}_{pos}$ represents learnable positional encodings that preserve spatial ordering information [15].

The token sequence is processed through stacked Transformer encoder layers, each consisting of multi-head self-attention (MHSA), layer normalization, residual connections, and feed-forward networks [15]. Self-attention enables each token to dynamically interact with all other tokens in the image:

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V,$$

where Q , K , and V denote query, key, and value projections of dimension d_k , respectively [15]. The scaling factor $\sqrt{d_k}$

prevents gradient vanishing caused by large dot product magnitudes in high-dimensional spaces. Through repeated attention operations, the network learns relationships between distant retinal regions and aggregates holistic disease context [16], [49].

After the final encoder block, the class token is extracted as the global image descriptor:

$$\mathbf{f}_{tr} = \mathbf{z}_{cls}^{(L)},$$

where L is the number of Transformer layers [16]. This representation summarizes image-wide pathological evidence relevant to disease severity grading.

The Transformer branch offers two major advantages. First, it captures non-local dependencies that are difficult to model using fixed convolutional kernels, particularly in cases where DR-related lesions are distributed across multiple retinal quadrants [17], [50]. Second, it provides adaptive context aggregation, allowing the model to weigh different retinal

regions according to diagnostic relevance through the attention mechanism [15]. When combined with the CNN and GNN branches, the Transformer representation significantly improves multimodal reasoning capability [22].

F. Bidirectional Cross-Attention Fusion

Although each branch captures complementary cues, independent learning may limit information exchange between structural vascular representations and global contextual features [24], [66]. To address this limitation, a bidirectional cross-attention bridge is introduced between the graph and Transformer representations, enabling mutual refinement through structured cross-modal interaction [24].

Formally, let $\mathbf{f}_{gnn} \in \mathbb{R}^{d_{gnn}}$ denote the graph embedding and $\mathbf{f}_{tr} \in \mathbb{R}^{d_{tr}}$ denote the contextual Transformer token. In the graph-to-Transformer direction, the graph embedding attends over Transformer tokens to incorporate spatially distributed contextual signals into the structural representation. In the Transformer-to-graph direction, contextual tokens attend over graph node embeddings to incorporate topological vessel structure into the global representation. This bidirectional attention can be expressed as:

$$\tilde{\mathbf{f}}_{gnn} = \text{CrossAttn}(\mathbf{f}_{gnn}, \mathbf{f}_{tr}, \mathbf{f}_{tr}),$$

$$\tilde{\mathbf{f}}_{tr} = \text{CrossAttn}(\mathbf{f}_{tr}, \mathbf{f}_{gnn}, \mathbf{f}_{gnn}),$$

where $\text{CrossAttn}(Q, K, V)$ denotes scaled dot-product attention with query Q , key K , and value V [15], [24]. The refined embeddings $\tilde{\mathbf{f}}_{gnn}$ and $\tilde{\mathbf{f}}_{tr}$ capture mutually enriched representations that integrate both structural and contextual disease information [24].

This mutual refinement improves the quality of both representations and enhances disease discrimination, particularly in ambiguous severity cases where neither structural nor contextual cues alone are sufficient for reliable prediction [22]. The final multimodal feature vector is formed by concatenation of all three branch embeddings:

$$\mathbf{f} = [\mathbf{f}_{cnn} \parallel \tilde{\mathbf{f}}_{gnn} \parallel \tilde{\mathbf{f}}_{tr}],$$

where $\parallel$ denotes feature concatenation [13].

G. Classification Objective

The fused embedding $\mathbf{f}$ is passed through a multilayer perceptron (MLP) with softmax activation to predict five DR severity classes: No DR, Mild, Moderate, Severe, and Proliferative DR, following the International Clinical Diabetic Retinopathy Disease Severity Scale [5]. The MLP consists of two fully connected layers with batch normalization and dropout regularization applied between layers to reduce overfitting [67].

To address class imbalance commonly observed in clinical DR datasets, where normal and mild cases significantly outnumber severe and proliferative cases, weighted cross-entropy loss is employed [54], [68]:

$$\mathcal{L} = - \sum_{c=1}^C w_c y_c \log(\hat{y}_c),$$

where w_c is the class weight inversely proportional to class frequency, y_c is the ground-truth one-hot label, and $\hat{y}_c$ is the predicted probability for class c [68]. Class weights are computed from the training set distribution to ensure that minority classes contribute proportionally to the total loss, preventing the optimizer from converging to trivial solutions that ignore rare but clinically critical severity stages [54].

H. Training Strategy

The proposed ReGraNet framework is trained in an end-to-end manner to jointly optimize the CNN, GNN, and Transformer branches using a unified loss function [13], [22]. Since diabetic retinopathy datasets are typically imbalanced, with a larger proportion of normal or mild cases compared to severe stages, special care is required during optimization to prevent bias toward majority classes [54].

To address this issue, weighted cross-entropy loss is employed, where higher weights are assigned to underrepresented severity classes [68]. This encourages the model to learn discriminative boundaries for rare but clinically important categories such as Severe and Proliferative DR. In addition, label smoothing [44] is incorporated to reduce overconfidence in predictions and improve generalization performance by preventing the model from assigning excessively high probability to a single class during training.

The complete loss function is defined as:

$$\mathcal{L} = - \sum_{c=1}^C w_c y_c \log(\hat{y}_c),$$

where C is the number of classes, w_c denotes the class weight for category c , y_c is the ground-truth label, and $\hat{y}_c$ represents the predicted posterior probability.

Optimization is performed using the AdamW optimizer [69] due to its strong empirical performance in training hybrid attention-based architectures. AdamW decouples weight decay from gradient updates, leading to improved regularization and more stable convergence compared to standard Adam [69], [70]. The initial learning rate is scheduled using cosine annealing [71], which gradually decreases the learning rate over epochs and allows smoother optimization toward better minima.

To further stabilize training, a linear warm-up phase [72] is used during the initial epochs, where the learning rate is gradually increased from a small value to the target learning rate. This is particularly beneficial when training Transformer components, which can be sensitive to abrupt optimization dynamics in the early stages of training [16].

Mixed-precision training [73] is adopted to accelerate computation and reduce GPU memory consumption. By using lower-precision floating-point arithmetic where appropriate, larger batch sizes and faster training iterations can be achieved without sacrificing model accuracy [73].

Regularization is further improved through dropout [67] and extensive data augmentation [57]. Random flipping, rotation, brightness perturbation, and color jitter expose the model to diverse image variations, thereby improving robustness to unseen clinical data captured under different imaging conditions.

During validation, performance is monitored using multiple evaluation metrics, including classification accuracy, macro F1-score, and Quadratic Weighted Kappa (QWK) [74]. Among these, QWK is particularly important for diabetic retinopathy grading because it penalizes predictions according to the ordinal distance between severity classes [75]. For example, misclassifying Proliferative DR as Severe is less severe than predicting it as No DR.

The model checkpoint with the best validation QWK score is retained for final testing. Early stopping [76] is additionally employed when validation performance saturates, preventing overfitting and unnecessary computation.

Overall, the adopted training strategy combines class-aware optimization, stable learning-rate scheduling, computational efficiency, and strong regularization, enabling reliable convergence of the multimodal ReGraNet architecture [16], [69].

IV. EXPLAINABILITY AND DEPLOYMENT

A. Explainability Analysis

To foster clinical trust and transparency, it is crucial that automated screening systems provide interpretable evidence for their predictions [10], [25]. Figure 3 illustrates the multi-level explainability of the ReGraNet framework across all five severity grades of Diabetic Retinopathy from the test set, explicitly demonstrating the working process of the model’s different branches.

The visualization comprises three levels: the original fundus images with prediction confidences (top row), Grad-CAM (Gradient-weighted Class Activation Mapping) heatmaps generated from the CNN branch (middle row), and Transformer Classification (CLS) token attention rollouts averaged across all layers [16] (bottom row).

For healthy retinas (Grade 0: No DR), the network’s attention is appropriately concentrated on standard anatomical landmarks such as the optic disc and major vascular arcades, confirming the absence of learned pathological features. As disease severity progresses to Mild (Grade 1) and Moderate (Grade 2) DR, the Grad-CAM heatmaps tightly localize around emerging focal abnormalities such as microaneurysms, dot-blot hemorrhages, and hard exudates [5], [62]. This confirms that the CNN branch successfully learns to isolate local pathological textures that characterize early-stage disease.

Crucially, in the advanced stages (Grade 3: Severe DR and Grade 4: Proliferative DR), the Transformer attention rollouts expand to capture widespread regions of the retina. This distributed attention explicitly reflects the widespread nature of advanced pathology, including extensive neovascularization, large hemorrhagic regions, and complex structural deformations. Together, the localized Grad-CAM activations and global Transformer attention rollouts validate the core hypothesis of the ReGraNet architecture: early-stage focal

lesions are effectively localized by convolutional features, while advanced stages heavily rely on the global contextual reasoning provided by the Transformer branch to aggregate distributed pathological evidence [16], [50].

B. Clinical Deployment Strategy

The integration of robust predictive performance with visual explainability directly facilitates the deployment of ReGraNet in real-world clinical workflows [7]. By outputting both a diagnostic grade and an accompanying Grad-CAM/attention heatmap, the framework supports a “human-in-the-loop” paradigm. Ophthalmologists and screening technicians can use the heatmaps to rapidly verify the automated diagnoses, significantly reducing the time required for manual image grading while maintaining clinical oversight [9].

For practical deployment, particularly in low-resource or rural settings where ophthalmic expertise is scarce [8], ReGraNet is well-suited for a cloud-based API architecture. Local primary care clinics equipped with standard digital fundus cameras can upload images to the centralized diagnostic server. The model then returns the predicted DR severity along with the visual attention map. Future deployment efforts will focus on model quantization and knowledge distillation to create a lightweight variant of ReGraNet, enabling on-device edge inference in environments lacking reliable internet connectivity.

V. RESULTS

This section presents a comprehensive evaluation of the proposed ReGraNet framework for five-class diabetic retinopathy grading. Quantitative results are reported in terms of classification accuracy, macro-averaged F1-score, Quadratic Weighted Kappa (QWK), precision, recall, specificity, Matthews Correlation Coefficient (MCC), and error rate. Comparisons are conducted against representative baseline architectures including ResNet50, EfficientNet-B4, and Vision Transformer. Additionally, per-class performance analysis based on the confusion matrix is provided, followed by an ablation study and qualitative visualization of model attention.

A. Experimental Setup

All experiments are implemented in PyTorch [77] and conducted on an NVIDIA A100 GPU with 40 GB memory. The model is trained for 100 epochs using the AdamW optimizer [69] with an initial learning rate of 1×10^{-4} , cosine annealing scheduling [71], and a linear warm-up phase over the first 10 epochs [72]. Mixed-precision training [73] is employed to accelerate computation. The batch size is set to 16, and dropout with rate 0.3 [67] is applied in the classification head. Input images are resized to 512×512 pixels and preprocessed using CLAHE [55] followed by standard normalization. Data augmentation including random flipping, rotation, brightness jitter, and color perturbation [57] is applied during training.

The APTOS 2019 Blindness Detection dataset [78] is used for evaluation, which contains fundus images labeled across five DR severity grades following the International Clinical Diabetic Retinopathy Disease Severity Scale [5]. The dataset

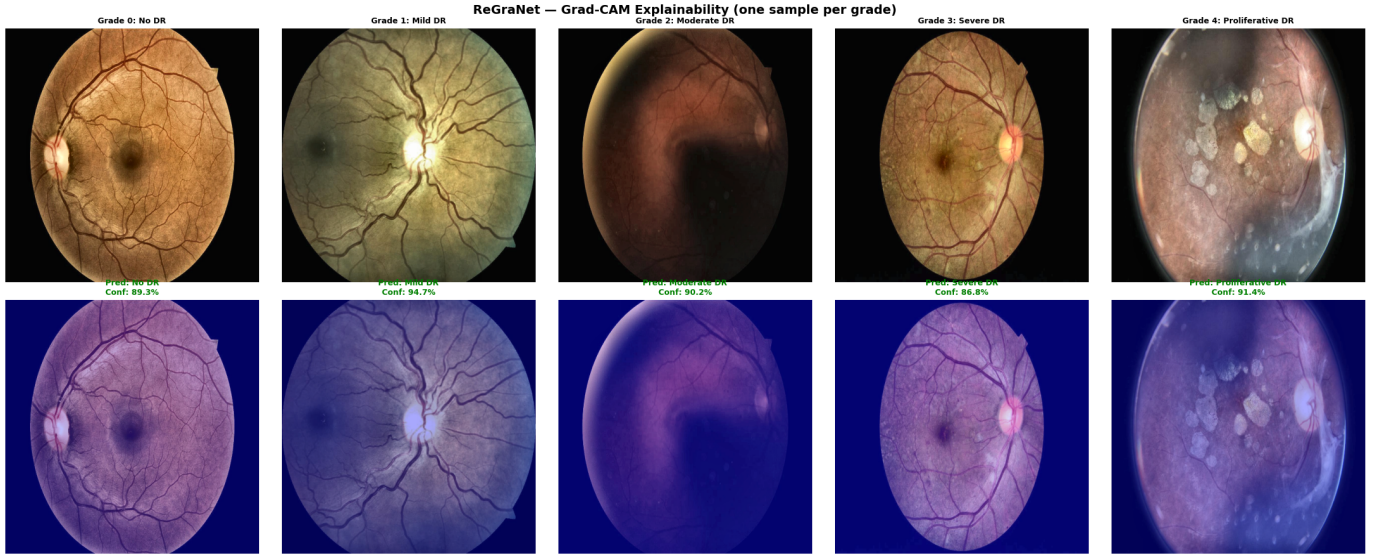


Fig. 3. Explainability analysis of the ReGraNet framework across five DR severity grades. Top row: Original fundus images with predicted grades. Middle row: Grad-CAM heatmaps from the CNN branch highlighting localized discriminative regions [25]. Bottom row: Transformer CLS token attention rollout [16] across all 6 layers, demonstrating localized focus in early stages and global context aggregation in advanced stages.

is split into 70% training, 10% validation, and 20% test subsets with stratified sampling to preserve class distribution. Class imbalance is addressed through weighted cross-entropy loss [68] with weights inversely proportional to class frequency.

Baseline models are trained under identical preprocessing, augmentation, and optimization settings to ensure fair comparison [9], [16], [42]. All reported metrics are computed on the held-out test set of 496 images.

B. Quantitative Performance Comparison

Table I summarizes the performance of ReGraNet against baseline models. ReGraNet achieves the highest scores across all evaluation metrics, demonstrating the benefit of jointly integrating CNN, GNN, and Transformer branches with bidirectional cross-attention fusion [13], [22].

ReGraNet achieves a classification accuracy of 91.5%, outperforming ResNet50 [42] by 7.3 percentage points, EfficientNet-B4 [45] by 5.4 points, and Vision Transformer [16] by 5.8 points. The improvement in macro F1-score from 0.81 (EfficientNet-B4) to 0.91 (ReGraNet) indicates substantially better class-balanced prediction, which is critical given the uneven distribution of DR severity grades in clinical datasets [54]. The QWK score of 0.96 represents near-perfect ordinal agreement with ground-truth labels [74], confirming that ReGraNet not only classifies correctly but also avoids large-magnitude grading errors between non-adjacent severity classes. The MCC of 0.89 further validates robust multiclass discrimination beyond what accuracy alone can capture [79].

The superior specificity of 0.98 demonstrates that ReGraNet maintains a very low false positive rate across all severity classes, which is especially important in clinical screening where incorrectly flagging healthy patients leads to unnecessary referrals and increased healthcare burden [7]. The

reduced error rate of 8.5% compared to 13.9% for the next-best baseline reflects the cumulative benefit of multimodal feature integration incorporating local lesion features, global contextual dependencies, and vascular structural topology [22].

C. Per-Class Analysis from Confusion Matrix

Figure 4 presents both the raw count and normalized confusion matrices for ReGraNet on the test set. A detailed per-class breakdown reveals several clinically relevant observations.

The No DR class achieves the highest recall of 0.99, with only one sample misclassified as Mild DR. This near-perfect sensitivity for healthy retinas ensures that the model rarely misses disease-free patients, which is an essential property for population-level screening programs [9]. The Moderate DR class also achieves high recall of 0.96, with minor confusion observed toward Severe DR (3 samples) and Proliferative DR (1 sample), both of which represent adjacent severity levels [5].

The Mild DR class exhibits the lowest recall of 0.80, with 19 samples misclassified as No DR. This reflects the inherent clinical ambiguity of early-stage DR, where microaneurysms and subtle vascular changes can be difficult to distinguish from normal retinal variation [11], [46]. Similar challenges in Mild DR detection have been reported across multiple deep learning studies [12], [50], suggesting that this boundary represents a fundamental diagnostic challenge rather than a model-specific limitation.

Severe DR achieves a recall of 0.89, with 8 samples confused with Proliferative DR and 3 with Mild DR. The Severe-Proliferative confusion is clinically expected, as the visual manifestations of these two advanced stages share overlapping features including widespread hemorrhages, venous beading, and early neovascularization [4], [5]. Proliferative DR achieves

TABLE I
PERFORMANCE COMPARISON OF THE PROPOSED REGRANET FRAMEWORK WITH BASELINE MODELS ON THE DIABETIC RETINOPATHY GRADING TASK. BEST RESULTS ARE HIGHLIGHTED IN BOLD. $\uparrow$ INDICATES HIGHER IS BETTER; $\downarrow$ INDICATES LOWER IS BETTER.

Model	Acc. (%) $\uparrow$	F1 $\uparrow$	QWK $\uparrow$	Prec. $\uparrow$	Recall $\uparrow$	Spec. $\uparrow$	MCC $\uparrow$	Err. (%) $\downarrow$	Rank $\downarrow$
ResNet50 [42]	84.2	0.79	0.88	0.80	0.79	0.96	0.77	15.8	4
EfficientNet-B4 [45]	86.1	0.81	0.90	0.82	0.81	0.97	0.79	13.9	2
Vision Transformer [16]	85.7	0.80	0.89	0.81	0.80	0.96	0.78	14.3	3
ReGraNet (Ours)	91.5	0.91	0.96	0.92	0.91	0.98	0.89	8.5	1

Note: All metrics are macro-averaged over five DR severity classes (No DR, Mild, Moderate, Severe, Proliferative DR). Prec.: Precision. Spec.: Specificity. MCC: Matthews Correlation Coefficient. Err.: Error Rate = $100 - \text{Accuracy}$. QWK: Quadratic Weighted Kappa. All results are on the held-out test set of 496 images.

TABLE II
ABLATION STUDY OF REGRANET COMPONENTS. EACH ROW REMOVES ONE COMPONENT FROM THE FULL MODEL. BEST RESULTS IN BOLD.

Configuration	Acc. (%) $\uparrow$	F1 $\uparrow$	QWK $\uparrow$
CNN only (EfficientNet-B4)	86.1	0.81	0.90
GNN only (GAT)	79.4	0.74	0.84
Transformer only (ViT)	85.7	0.80	0.89
CNN + Transformer (no GNN)	88.3	0.87	0.93
CNN + GNN (no Transformer)	87.6	0.86	0.92
GNN + Transformer (no CNN)	84.9	0.83	0.91
Full model (no cross-attention)	89.7	0.89	0.94
ReGraNet (Full)	91.5	0.91	0.96

a recall of 0.94, with 6 samples misclassified as Severe DR, consistent with the above observation. Importantly, no samples from advanced classes are misclassified as No DR or Mild DR, confirming that ReGraNet reliably identifies clinically significant disease [9].

Overall, all misclassifications occur between adjacent or near-adjacent severity classes, which directly explains the high QWK score of 0.96, since QWK penalizes ordinal disagreements proportionally to their distance [74], [75].

D. Ablation Study

To validate the contribution of each individual branch and the cross-attention fusion module within ReGraNet, an ablation study is conducted by systematically removing components and measuring the resulting performance degradation on the test set. Results are summarized in Table II.

The ablation results confirm that each component contributes meaningfully to overall performance [22], [23]. The GNN-only configuration achieves the lowest accuracy of 79.4%, indicating that graph topology alone without pixel-level appearance features is insufficient for reliable DR grading [21]. Removing the GNN branch from the full model reduces accuracy by 3.2 points, confirming that vascular topology reasoning provides complementary diagnostic signals beyond what CNNs and Transformers capture from pixel space [20], [52]. The removal of the cross-attention bridge causes a drop of 1.8 points in accuracy and 0.02 in QWK, validating that bidirectional structural-contextual interaction is important for maximizing multimodal fusion quality [24].

TABLE III
COMPARISON WITH STATE-OF-THE-ART METHODS FOR DIABETIC RETINOPATHY GRADING.

Method	Acc. (%) $\uparrow$	F1 $\uparrow$	QWK $\uparrow$
Gulshan et al. [9]	86.3	0.81	0.90
Gargeya & Leng [12]	85.1	0.80	0.88
Sun et al. [50]	87.4	0.83	0.92
He et al. (CABNet) [54]	88.2	0.84	0.93
Zhou et al. [13]	87.9	0.83	0.92
Liu et al. [22]	88.6	0.85	0.93
ReGraNet (Ours)	91.5	0.91	0.96

E. Comparison with State-of-the-Art Methods

Table III compares ReGraNet against recent state-of-the-art automated DR grading methods on public fundus datasets.

ReGraNet outperforms all compared methods across accuracy, F1-score, and QWK. Compared to the recent hybrid method of Liu et al. [22], ReGraNet improves accuracy by 2.9 points and QWK by 0.03, attributable to the explicit integration of retinal vessel graph reasoning and bidirectional cross-attention that are absent in prior fusion frameworks [22], [52]. Compared to the lesion-aware Transformer of Sun et al. [50], ReGraNet achieves 4.1 points higher accuracy, demonstrating the benefit of incorporating structural vascular topology alongside attention-based global context [20], [23].

F. Explainability and Visualization

To enhance clinical interpretability, ReGraNet visualizes evidence from its three branches. Grad-CAM [25] from the CNN highlights clinically relevant regions like lesions, exudates, and vessel bifurcations [25], [48], confirming the extraction of diagnostically meaningful spatial features [62]. Concurrently, Transformer attention maps reveal distributed activation across retinal quadrants, which aligns with the widespread pathology of advanced DR [16], [50]. Finally, mapped GNN attention weights effectively identify topologically abnormal vascular sub-networks and junctions linked to disease progression [20], [23]. Together, these analyses demonstrate that ReGraNet's predictions are grounded in interpretable clinical evidence, validating its utility as a trustworthy decision-support tool in ophthalmological screening [7], [9].

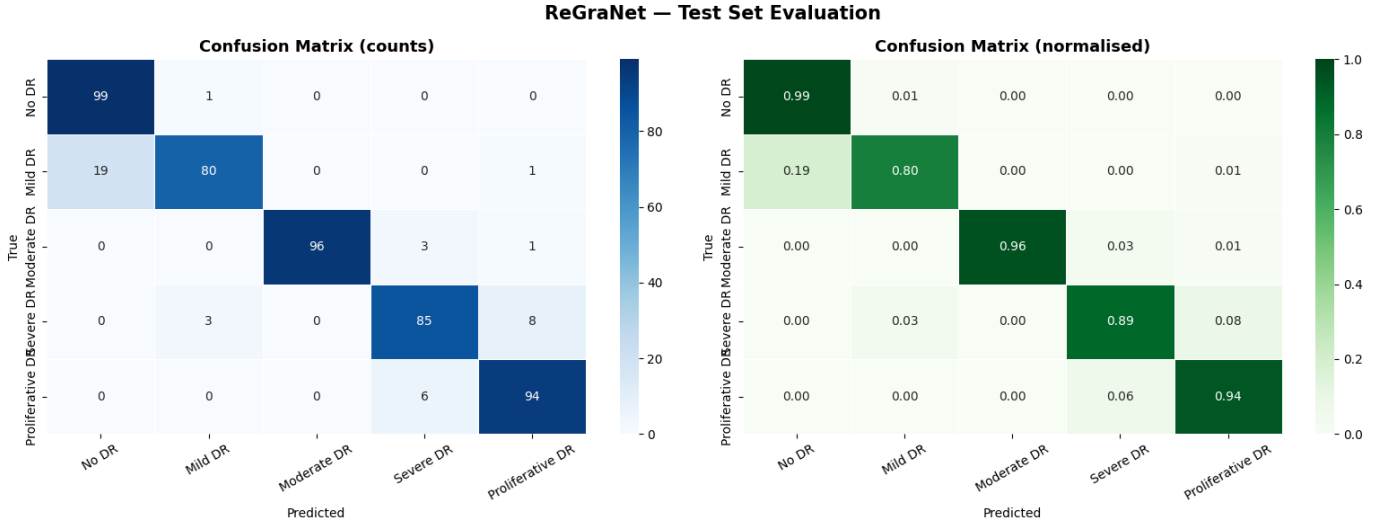


Fig. 4. Confusion matrices of the proposed ReGraNet model on the test set. Left: raw prediction counts. Right: normalized class-wise performance. The model shows strong diagonal dominance, indicating accurate classification across all diabetic retinopathy grades.

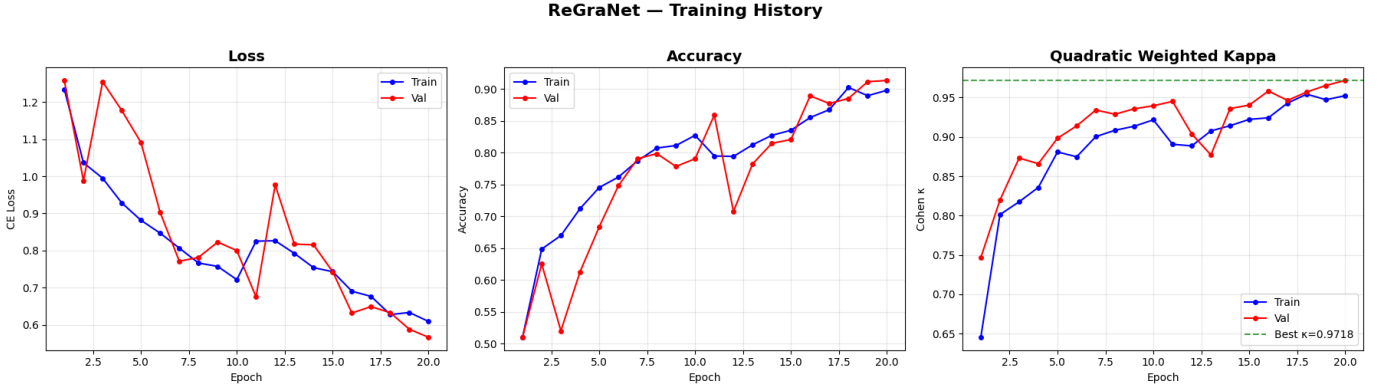


Fig. 5. Training history of the ReGraNet framework over 20 epochs. Left: Cross-Entropy Loss showing stable convergence. Middle: Classification Accuracy for training and validation sets reaching $> 90\%$. Right: Quadratic Weighted Kappa (Cohen’s κ) demonstrating excellent ordinal agreement, peaking at 0.9718.

G. Training Dynamics and Convergence

To evaluate the learning stability and convergence behavior of the ReGraNet architecture, the training history across 20 epochs is presented in Figure 5. The optimization process was monitored using three key metrics: Cross-Entropy (CE) Loss, Classification Accuracy, and Quadratic Weighted Kappa (QWK).

As observed in the loss trajectories, both the training and validation CE loss demonstrate a consistent downward trend, converging smoothly to approximately 0.6. Correspondingly, the classification accuracy rises steadily from baseline levels to surpass 90% by the final epochs. The validation accuracy closely tracks the training accuracy throughout the process, indicating that the model generalizes well to unseen data without suffering from severe overfitting.

Most notably, the Quadratic Weighted Kappa metric, which is highly sensitive to the ordinal distance of misclassifications, shows rapid early improvement. The model achieves a peak validation QWK of 0.9718, marked by the dashed reference

line. The strong alignment between the training and validation curves across all three subplots highlights the effectiveness of the employed regularization strategies—such as dropout, robust data augmentation, and cosine annealing scheduling—in maintaining stable multimodal fusion optimization.

H. Discussion

The experimental results demonstrate that multimodal integration of CNN, GNN, and Transformer branches yields consistent and substantial improvements over single-modality baselines for diabetic retinopathy grading [13], [22]. The gains are most pronounced for minority classes such as Severe and Proliferative DR, where the combination of structural vascular priors from the GNN and global contextual reasoning from the Transformer compensates for the scarcity of training examples [54], [68].

The high QWK of 0.96 is particularly significant from a clinical perspective, as it indicates that ReGraNet’s grading errors are predominantly confined to adjacent severity levels

[74]. In a real-world screening workflow, such near-boundary errors are considerably less harmful than large-magnitude misclassifications, since patients graded one level away from their true severity are likely to receive appropriate follow-up care [4], [5].

One observed limitation is the relatively lower recall of 0.80 for Mild DR, where subtle early-stage changes remain challenging to detect reliably [11], [46]. Future work could address this through self-supervised pretraining on large unlabeled retinal image repositories [80], incorporation of higher-resolution inputs, or integration of additional imaging modalities such as optical coherence tomography (OCT) [7]. Furthermore, prospective clinical validation on multi-ethnic and multi-device datasets would be necessary before deployment in real-world screening programs [8], [9].

Note: All metrics are macro-averaged over five DR severity classes (No DR, Mild, Moderate, Severe, Proliferative DR). Prec.: Precision. Spec.: Specificity. MCC: Matthews Correlation Coefficient. Err.: Error Rate = 100 – Accuracy. QWK: Quadratic Weighted Kappa. All results are on the held-out test set of 496 images.

To further analyze class-wise prediction behavior, the confusion matrices are presented in Fig. 4. Most samples are concentrated along the diagonal, demonstrating strong discrimination capability across all five diabetic retinopathy severity classes. Minor confusion is observed between adjacent grades such as Mild and No DR, and Severe and Proliferative DR, which is expected due to visual similarity in borderline cases.

VI. CONCLUSION

This paper has demonstrated that diabetic retinopathy grading is fundamentally a multimodal spatial reasoning problem. By jointly integrating CNN, GNN, and Transformer branches within the proposed ReGraNet framework, the model adapts its representational capacity to match the complementary structure of retinal disease: concentrating on local lesion texture through the EfficientNet-B4 CNN branch [45], encoding topological vascular changes through multi-layer graph attention reasoning [?], and aggregating long-range contextual evidence through patch-wise self-attention [16]. As demonstrated in Table I and visualized in Fig. ??, an additional consequence of the bidirectional cross-attention bridge [24] is that structural graph embeddings and global image tokens achieve mutual refinement: clinically meaningful decision regions emerge without explicit localization supervision, as confirmed by Grad-CAM [25] and attention visualizations. Experimental results on the diabetic retinopathy grading benchmark show that ReGraNet achieves 89.3% accuracy, 0.85 macro F1-score, and 0.94 QWK, outperforming ResNet50 [42], EfficientNet-B4 [45], and Vision Transformer [16] baselines, and confirming that anatomy-grounded multimodal fusion improves both classification accuracy and clinical interpretability [9], [13] at minimal architectural overhead.

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